Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/05/2019
PI Name:	Ju, Yiguang Ph.D.		
Project Title:	Quantitative Studies of Cool Flame Transitions at Radiation/Stretch Extinction Using Counterflow Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08540-5263	**Congressional District:**	12
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-A: Use of the NASA Physical Sciences Informatics System – Appendix A
Start Date:	08/01/2016	**End Date:**	07/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2020 per NSSC information (Ed., 11/12/19) NOTE: Extended to 7/31/2019 per NSSC information (Ed., 5/7/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reuter, Christopher M.S. (Princeton University)		
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Performance Goal Text:			

Task Description:	Cool flames have significant implications for the control of engine knocking, ground and space fire safety, and combustion emissions. The recent International Space Station (ISS) combustion science experiments on flame extinguishment discovered the existence of cool flames for both large single droplets and droplet arrays after radiative extinction. Numerical simulations of droplet combustion also provide an explanation for the transition of hot flame to cool flame after radiative extinction. However, it remains unclear at what conditions the hot flames transfer to cool flames and how the cool flame flammable region is related to hot flame flammability limit and extinction limit. It is also not clear how oxygen concentration, dilution, pressure, diffusion, and convection transport processes affect the cool flame transition. More importantly, due to limits of space and payload, the ISS experiments did not provide detailed speciation and temperature distributions of cool flame structures for quantitative validation of kinetics and radiation model. By using the Flame Extinguishment (FLEX) experimental data from the Physical Sciences Informatics (PSI) system of combustion science, we propose to conduct new simulations and ground-based experiments using ozone-sensitized counterflow flames to * develop a new diagram of the flammability limits of diffusional cool flames affected by radiation, diffusion, and convection ; * bridge the knowledge gap between the flammability limits and extinction limits of hot flames and cool flames ; * predict and answer the question at what conditions the transition from hot flame extinction to cool flames is possible and verify the phenomena observed on the ISS flame extinguishment experiments ; * obtain new, quantitative data of extinction limit, flame temperature, and species distribution for the validation of kinetic and radiation models involving cool flames ; * extend the cool flame experiments from the original ISS droplet experiments to a more general ground-based combustion system involving radiation, diffusion, convection, and low-temperature chemistry while maintaining a one-dimensional geometry. The proposed new ground-based experiments and data analysis are complimentary to and not directly linked with the science objectives from the original ISS combustion science investigation, and will significantly advance the understanding of near-limit cool flame dynamics and contribute to the future control of engine knocking, ground and space fire safety, and combustion emissions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An experimentally validated dimethyl ether (DME) and methane cool flame kinetic model was developed. The results showed that the existing kinetic model significantly over-predicts the cool flame extinction limits. However, by targeting the exothermic reactions of cool flames using the cool flame extinction limits, an updated chemical kinetic model for dimethyl ether/methane mixtures is developed and validated. This study shows the value of the ozone-assisted counterflow cool flame platform in examining the key reactions contributing to the heat release rate in cool flames. Diffusion warm flames of n-alkanes were successfully established. The warm flame of n-alkanes is dramatically influenced by the reactant concentrations and mixture temperatures. The primary cool flame stage is favored at lower temperatures, but the secondary warm flame stage is favored at elevated temperatures. Moreover, near extinction, the multistage warm flame collapses to a primary oxidation zone, governed by peroxy chemistry, whereas the secondary oxidation stage, governed by intermediate chemistry, is dominant near autoignition conditions. The results show that warm flames serve as a critical bridge between cool flames and hot flames. The counterflow cool flame extinction provides a good validation target and explanation for the NASA droplet cool flame extinction limit. In addition, the results benefit the development of efficient and low mission advanced engines for transportation.
Task Progress:	An experimentally validated dimethyl ether (DME) and methane cool flame kinetic model was developed. The results showed that the existing kinetic model significantly over-predicts the cool flame extinction limits. However, by targeting the exothermic reactions of cool flames using the cool flame extinction limits, an updated chemical kinetic model for dimethyl ether/methane mixtures is developed and validated. This study shows the value of the ozone-assisted counterflow cool flame platform in examining the key reactions contributing to the heat release rate in cool flames. Diffusion warm flames of n-alkanes were successfully established. The warm flame of n-alkanes is dramatically influenced by the reactant concentrations and mixture temperatures. The primary cool flame stage is favored at lower temperatures, but the secondary warm flame stage is favored at elevated temperatures. Moreover, near extinction, the multistage warm flame collapses to a primary oxidation zone, governed by peroxy chemistry, whereas the secondary oxidation stage, governed by intermediate chemistry, is dominant near autoignition conditions. The results show that warm flames serve as a critical bridge between cool flames and hot flames. The counterflow cool flame extinction provides a good validation target and explanation for the NASA droplet cool flame extinction limit. In addition, the results benefit the development of efficient and low mission advanced engines for transportation. Progress We made two main progresses in Year 3 of our research: (1) An experimentally validated dimethyl ether (DME) and methane cool flame kinetic model was developed; and (2) we established successfully diffusion warm flames of n-alkanes. In this first task, the extinction strain rates of hot flames, cool flames, and ozone-assisted cool flames have been measured in a counterflow burner. The reactivity of the hot flames is promoted by methane addition on a fuel mass fraction basis. However, the reactivity of cool flames (both ozone-less and ozone-assisted) is inhibited by methane, even when exchanged for inert nitrogen. Moreover, the ozone-assisted cool flames are found to reduce the sensitivity of cool flame extinction to the fuel concentration. A rate constant sensitivity analysis reveals that the CH3OCH3 + OH = CH3OCH2 + H2O and CH2OCH2O2H = CH2O + CH2O + OH reactions are disproportionally important to cool flames over homogeneous reactors, especially when considering their exothermic nature. These reactions are then modified (along with other reactions) based on the present experimental cool flame extinction data as well as reaction rates in the literature to produce an updated chemical kinetic model. The updated model is then validated using a combination of homogeneous reactors and flames at both high and low temperatures. Due to the increase in the rate constant of the CH3OCH3 + OH = CH3OCH2 + H2O reaction, the current model successfully predicts the cool flame extinction strain rate within ±20 s-1 of the experiments for DME/CH4 mixtures. The other experimental validation targets see differences between the original model and the updated model which are on the same order as the experimental uncertainties. A path flux analysis performed for cool flames reveals that the addition of methane

	scavenges OH and HO2 radicals, which is the reason behind its inhibition at low temperatures. This study shows the value of the ozone-assisted counterflow cool flame platform in chemical kinetic model development through its coupling of low-temperature chemistry, transport, and heat release. Specifically, it acts to constrain reactions that contribute significantly to the heat release and are subject to radical diffusion. In the second task, we demonstrate experimentally, for the first time, the existence of multistage low-temperature diffusion flames in the counterflow configuration at terrestrial conditions. Such flames are defined by two distinct sequential oxidation zones, where the parent fuel peroxy chemistry governs the primary stage and the subsequent oxidation of the intermediates produced sustains the secondary stage, for sufficiently large Damköhler number. Specifically, the intermediates produced through the degenerate branching pathway and the decomposition of the alkylketohydroperoxide oxidize and partially replenish the radical pool and consume the fuel in the secondary stage. The reaction zone structure of multistage flames is dramatically influenced by the fuel inlet temperature, where the primary oxidation stage is dominant at lower temperatures while the secondary stage is dominant at elevated temperatures. However, as both structures are coupled, the reactivity of the multistage flame is dictated by the primary stage, except when the branching and heat release are supported externally, as in the case of ozone-doping. The results suggest that such multistage character likely manifests in flow fields with sufficiently long residence times, such as microgravity environments and highly turbulent flow fields. The present work contributes directly to the understanding of low-temperature flames. Publications (see also Bibliography section below): Reuter, C.B., Zhang, R., Yehia, O.R., Rezgui, Y. and Ju, Y., 2018. Counterflow flame experiments and chemical kinetic modeling of dimethyl ether/methane mixtures. Combustion and Flame, 196, pp. 1-10. Reuter, C.B., Katta, V.R., Yehia, O.R. and Ju, Y., 2019. Transient interactions between a premixed double flame and a vortex. Proceedings of the Combustion Institute, 37(2), pp. 1851-1859. Yehia, O.R., Reuter, C.B. and Ju, Y., 2018. Low-temperature multistage warm diffusion flames. Combustion and Flame, 195, pp. 63-74. Yehia, O.R., Reuter, C.B. and Ju, Y., 2019. On the chemical characteristics and dynamics of n-alkane low-temperature multistage diffusion flames. Proceedings of the Combustion Institute, 37(2), pp. 1717-1724.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Reuter CB, Zhang R, Yehia OR, Rezgui Y, Ju Y. "Counterflow flame experiments and chemical kinetic modeling of dimethyl ether/methane mixtures." Combustion and Flame. 2018 Oct;196:1-10. https://doi.org/10.1016/j.combustflame.2018.06.004 , Oct-2018
Articles in Peer-reviewed Journals	Reuter CB, Katta VR, Yehia OR, Ju Y. "Transient interactions between a premixed double flame and a vortex." Proceedings of the Combustion Institute. 2019;37(2):1851-9. https://doi.org/10.1016/j.proci.2018.07.050 , Jan-2019
Articles in Peer-reviewed Journals	Yehia OR, Reuter CB, Ju Y. "Low-temperature multistage warm diffusion flames." Combustion and Flame. 2018 Sep;195:63-74. https://doi.org/10.1016/j.combustflame.2018.03.013 , Sep-2018
Articles in Peer-reviewed Journals	Yehia OR, Reuter CB, Ju Y. "On the chemical characteristics and dynamics of n-alkane low-temperature multistage diffusion flames." Proceedings of the Combustion Institute. 2019 Jan;37(2):1717-24. https://doi.org/10.1016/j.proci.2018.06.161 , Jan-2019
Articles in Peer-reviewed Journals	Lin E, Reuter CB, Ju Y. "Dynamics and burning limits of near-limit hot, warm, and cool diffusion flames of dimethyl ether at elevated pressures." Proceedings of the Combustion Institute. 2019;37(2):1791-8. https://doi.org/10.1016/j.proci.2018.05.082 , Jan-2019